Program : **Diploma in Cloud Computing And Big Data**	
Course Code : **4277**	Course Title: **Web Application Development Lab**
Semester : **4**	Credits: **2.5**
Course Category: **Engineering Science**	
Periods per week: **4 (L: 1 T: 0 P: 3)**	Periods per semester: **60**

Course Objectives:

- To impart the design, development and implementation of dynamic web pages.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Knowledge in Computer Systems	1008	Introduction to IT Systems Lab	1
Basic Programming Knowledge		Problem Solving and Programming Programming in Python Lab	2
Basic Knowledge in Database		Database System Concepts & Database Lab	3

Course Outcomes:

On completion of the course student will be able to:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Construct interactive web pages using HTML	11	Applying
CO2	Prepare an HTML document using Cascaded Style Sheets, and JavaScript/jQuery	17	Applying
CO3	Experiment with PHP and MYSQL to construct Web applications	14	Applying

CO4	Design, develop and deploy(host) a web based application	12	Applying
	Lab Test	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	2	1				
CO3	2	2	1				
CO4	3	2	1	1			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Construct interactive web pages using HTML**		
M1.01	Overview of Website, domain name, web hosting, client-server communication	2	Understanding
M1.02	Demonstrate basic Syntax of HTML, Standard HTML Document Structure, Navigation	2	Applying
M1.03	Use basic Text Markup, Images, Hypertext Links, Lists, and Tables in HTML	3	Applying
M1.04	Illustrate HTMLtags for handling forms - form, input, text, select, radiobutton, check box, file upload, submit	4	Applying
CO2	**Prepare an HTML document using Cascaded Style Sheets, and JavaScript/jQuery**		

M2.01	Understand What is CSS, CSS Syntax, CSS Selectors Content style- element properties- font, list, background, border, color,	3	Applying
M2.02	CSS- Spacing- padding, margin, Indentation, letter spacing, word spacing, line height. Positioning, Animation	2	Applying
M2.03	Use bootstrap for designing an HTML page , CSS icons	2	
M2.04	Introduction to JavaScript and jQuery, Object Oriented concept , General Syntactic Characteristics such as primitives, operations, and expressions	3	Applying
M2.05	Construct Javascript programs for Screen Output Keyboard Input, Control Statements and input validation	3	
M2.06	Implement arrays and functions using javascript	2	
	Lab Test – I	2	
CO3	**Experiment with PHP and MYSQL to construct Web applications**		
M3.01	Demonstrate Primitives, Operations, and Expressions in PHP	3	Applying
M3.02	Construct PHP programs using Control Statements, Arrays, and Functions	4	Applying
M3.03	Illustrate MySQL database operations such as create and drop table/database, select, insert, delete and update.	4	Applying
M3.04	Implement Database in Web applications Using PHP and MYSQL	3	Applying
C04	**Design, develop and deploy(host) a web based application**		
M4.01	Setup development environment- IDE-LAMP/WAMP	2	Applying
M4.02	Conceptualise an interactive web application with database support- Design application, navigation, database and user interface	2	Applying
M4.03	Develop front end html pages with necessary JavaScript/jQuery and CSS	3	Applying

M4.04	Develop back-end PHP programs, Test the application, demonstrate and deploy	3	Applying
	Lab Test – II	2	

Sample Open Ended Projects

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. Avoid duplication of experiments between groups.)

1. Develop student registration form and display all the submitted data on another page
2. Develop a PHP program to read consumer details and display it on table format
3. Develop an interactive web application of your Institution
4. Develop an interactive web application for the Alumni association of your Institution
5. Design and develop a web magazine

Text / Reference

T/R	Book Title/Author
T1	Robert W Sebesta, Programming the World Wide Web, 7/e, Pearson Education Inc., 2014.

Online Resources

Sl No	Website Link
1	https://www.w3schools.com/
2	https://www.tutorialspoint.com/